

Capable — "turning frontier intelligence into frontier therapeutics"

A vertically integrated, ML-native drug design company claiming a 24-hour design-to-validated-data loop against a biotech industry standard of 4–12 weeks, with a lead program ("Ozempic for sleep") targeting short-sleeper pathways and a stated intent to enter the clinic this year.

VERDICT

EXTRAORDINARY CLAIMS — 3/10 ON THIS DECK

the most ambitious and the least evidenced of the three

THE TEAM

Genuinely exceptional

George Church and Luis de Lecea (co-discoverer of orexin) as advisors; Pentelute (MIT peptide synthesis); an unusually deep bench for a company <1 year old

THE CORE CLAIM

24h vs 4 weeks

a ~28× cycle-time compression; if true it is a genuinely new capability, and the deck shows no data supporting it

THE PROBLEM

No data in the deck

"4 picomolar potency," "world record to clinic," 34 indications — asserted, never shown

What the deck actually says

- **The thesis (strong):** frontier AI is bottlenecked by data, not intelligence — "you cannot hallucinate a drug into existence." Capable positions itself as the wet-lab compute layer where models get benchmarked by reality. This is a genuinely good insight and matches our own house view that physical ground truth is the scarce input.
- **The speed claim:** industry takes 4 weeks minimum per iteration because design (SF), synthesis (Hebei), cell testing (Shanghai) and mouse work (San Diego) are geographically fragmented — total 4–12 weeks and ~\$110k. Capable claims 24 hours by vertical integration plus in-house research agents, built in under six months.

- **Lead program (CAPABLE-002, "Ozempic for sleep"):** targets master regulators of wakefulness via short-sleeper gene biology; claims a stabilised, truncated peptide at "as low as 4 picomolar" potency achieved in 6 months. Beachhead markets narcolepsy (~\$4B) and hypersomnia (~\$1B), orphan/Fast-Track regulatory strategy, up to \$200k per patient per year, expanding to sleep apnea, depression, Alzheimer's, Parkinson's.
- **Program 3:** multi-agonists across several short-sleeper pathways, explicitly modelled on tirzepatide's GLP/GIP combination.
- **Roadmap:** "first drug to the clinic this year — that's a world record." A later slide lists 34 human-enhancement indications (cognitive enhancers, exercise mimetics, neuroprotection, body-odour elimination).

CAPABLE · 03

Why this scores low despite the team

The deck asks for belief in five extraordinary claims and evidences none of them: a 28× cycle-time compression; 4 pM potency in 6 months; clinic entry within the year; a peptide that safely compresses human sleep; and a platform that generalises to 34 indications. Any one would be a landmark. Presented together with no data, the aggregate reads as a vision document rather than a diligence artifact — and the deck is labelled "post-close," meaning the round is done and this is a narrative, not a solicitation.

The specific problems

- **"Ozempic for sleep" is a category error as a safety proposition.** Semaglutide's tolerability was established over two decades of GLP-1 biology in a disease population. Pharmacologically suppressing sleep need in healthy humans is a fundamentally different regulatory and ethical proposition; FDA has no accepted endpoint for "less sleep needed" in healthy subjects, which is why the deck's orphan-indication route (narcolepsy, hypersomnia) is the only viable path — and that path is a real, competitive market (Jazz, Takeda, Alkermes all active in orexin agonism) that the deck never mentions.
- **The orexin field has a hard clinical history.** Orexin-receptor agonism is not novel-and-unexplored; it is actively pursued, and de Lecea's presence signals credibility rather than freedom to operate. The competitive slide is absent entirely.
- **24 hours vs 4 weeks is a comparison against outsourced serial workflows.** Any integrated lab beats that benchmark; the honest comparison is against a well-run

internal discovery group, and the deck does not make it. Speed of iteration also does not compress the animal PK/tox and IND-enabling work that dominates time-to-clinic — which makes "clinic this year" the claim most likely to slip.

- **Program breadth is a warning, not a strength.** A company under a year old listing 34 target indications including body-odour elimination is signalling narrative reach, not focus.

CAPABLE · 04

Verdict and what would change it

3/10 on the deck as presented — not because the idea is bad (the data-bottleneck thesis is correct and the team is real), but because the deck contains no verifiable evidence for claims whose magnitude demands it, and because the safety-and-regulatory reality of the lead program is unaddressed. The very strong advisory bench raises rather than lowers the bar: with Church and de Lecea attached, data should be available to show.

What would move this materially

- The actual potency data for CAPABLE-002 — assay, replicates, counter-screens, selectivity across orexin receptor subtypes.
- One documented end-to-end 24-hour cycle: timestamps from design to validated readout, with the compound and the assay named.
- The regulatory plan in writing: pre-IND meeting status, the specific indication of first filing, and the endpoint FDA has agreed to consider.
- A competitive slide against Jazz/Takeda/Alkermes orexin programs, and freedom-to-operate on short-sleeper pathway IP (the DEC2/ADRB1 variants are academically encumbered).
- Cap table, round terms, burn, and runway — none appear in the deck.

Framing for the principal: of the three, Capable has the highest ceiling and the widest evidence gap. The correct move is not a pass on the company but a pass on *this artifact*: request the data room. If the 24-hour loop and the 4 pM potency are real and documented, the score changes substantially — those are checkable facts, and checkable facts are what the deck withholds.

Sources: CAPABLE — External (Post-close) Figma deck, 38 slides captured and read this session. No independent verification performed; all figures are deck-stated.